

Ayman Mahfuz

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EDUCATION

The University of Texas at Austin

Austin, TX

Bachelor of Science in Computer Science

May 2027

- **Concentration:** AI and ML — **GPA:** 3.5/4.0
- **Courses:** Data Structures, Algorithms, High Performance Computing, Natural Language Processing, Generative Visual Computing, Energy-Efficient Computing

EXPERIENCE

Arm

May 2025 – Aug 2026

Platform Validation Intern / Co-op – Software Engineering

- Reproduced silicon failures across **three CPU-platform validation phases** by building a production **Python Bayesian optimizer** for **stress-ng** configurations; owned development and deployment.
- Guided ML stress-test selection with **CPU performance counters**, optimizing **cache-miss rate**, **memory bandwidth**, and **stall rate** to identify high-stress configurations.
- Earned internal patent-filing approval and a **12-month** co-op extension; presented production results to SVP and director leadership.

UT Austin – Texas Robotics & AI Lab

Jan 2025 – Jul 2026

Undergraduate Research Assistant – Robotics RL & LLM Evals

- Built a seeded, multi-turn evaluation harness to test whether a coin-game RL advisor transfers to Persuade for Good donation dialogues; served three Qwen variants with vLLM on TACC GPUs and recorded splits, seeds, model IDs, manifests, and 54 full JSONL trajectories.
- Accelerated policy-training iterations **2–3x** by profiling **GPU** utilization and removing training-pipeline bottlenecks; scaled experiments to **5M+ episodes**.
- Deployed **15** hierarchical **reinforcement-learning** policies to **NAO robots** and built **C++/Python** training environments within a **400K+ line** codebase, contributing to the team's third-place finish at RoboCup 2025.

UT Austin – Center for Media Engagement

Aug 2023 – May 2025

Undergraduate Research Assistant – Software Engineering

- Built concurrent **Selenium/REST** ingestion workers with retry handling for a **250M+ row BigQuery dataset** containing **50M+ articles** and **70M+ comments**.
- Prevented duplicate experiment events and preserved round-robin assignments for **1,000+ participants** through deterministic assignment logic, event deduplication, and structured logging.

UT Austin – Oden Institute

Aug 2023 – Jan 2025

Undergraduate Research Assistant – ML Systems

- Improved MRI-segmentation Dice score by **12 percentage points** through **convolutional and transformer** model experiments with **five-fold validation** on **1,000+ scans** using **NVIDIA H100 GPUs**.

PROJECTS

HelmPM – Multi-Tenant Project Management Platform | *React, FastAPI, PostgreSQL*

- Built a multi-tenant platform managing **thousands of tickets across four organizations**, with a **React** frontend, **FastAPI** backend, **PostgreSQL** database, and automated **Linear/Jira** synchronization.
- Eliminated re-embedding of unchanged documents with **Merkle-tree indexing**, reducing API costs by **40%**; implemented **semantic, keyword, and exact-ID search** across documents and code.

Activation Checkpoint Integrity in PyTorch – Independent Research | *Python, PyTorch, CUDA*

- Built a PyTorch audit that pairs original and recomputed tensors during activation checkpointing, localizes silent value divergence, and gates the optimizer before mismatched gradients can mutate training state, under Professor Mootaz Elnozahy.

Energy Profiling of LLM & Diffusion Inference | *Python, PyTorch, vLLM, SLURM*

- Ran **577 GH200 experiments** across nine models and 55 configurations, measuring **24–32% lower LLM inference energy** with FP8/INT8 and no measurable diffusion savings; built a vLLM/SLURM harness instrumented with NVML/Zeus and analyzed results using Welch's t-tests, Bonferroni correction, and roofline analysis.

Transformer Pretraining & Post-Training Pipeline (253M Parameters) | *PyTorch, HPC*

- Pretrained a **253M-parameter transformer** on **600M tokens** with RoPE, RMSNorm, and SwiGLU; built resumable mixed-precision HPC training with checkpoint recovery, supervised fine-tuning, and DPO.

SKILLS

Languages: Python, C++, Java, TypeScript

Backend/Web/Data: React, FastAPI, Flask, PostgreSQL, BigQuery, REST APIs, Pandas, NumPy, AWS

ML/Systems: PyTorch, Transformers, vLLM, LoRA/PEFT, PPO/MAPPO, RLHF/DPO, SLURM, Docker, Linux, profiling, MCP, Harness Engineering, Zeus/NVML, HPC, Distributed Systems